

CUSTOMER CHURN PREDICTION

Retention Analytics Summary Report · Random Forest ML Model · Telecom Dataset

10,000	4,112	41.1%	77.59%	Rs. 24.5L
Total Customers	Churned	Churn Rate	Model ROC-AUC	Monthly Rev. at Risk

1. EXECUTIVE SUMMARY

A machine learning analysis of **10,000 telecom customers** was conducted to identify key churn drivers and build a predictive system capable of flagging at-risk customers before they leave. The overall churn rate stands at **41.1%** (4,112 churned out of 10,000), representing significant monthly revenue loss estimated at approximately **Rs. 24.5 Lakh/month**. A Random Forest classifier (300 trees) achieved a ROC-AUC of **77.59%** on the held-out test set. Combined retention interventions are projected to reduce churn by **25–30%**, saving approximately **Rs. 1.8–2.2 Crore annually**.

CHURN OVERVIEW

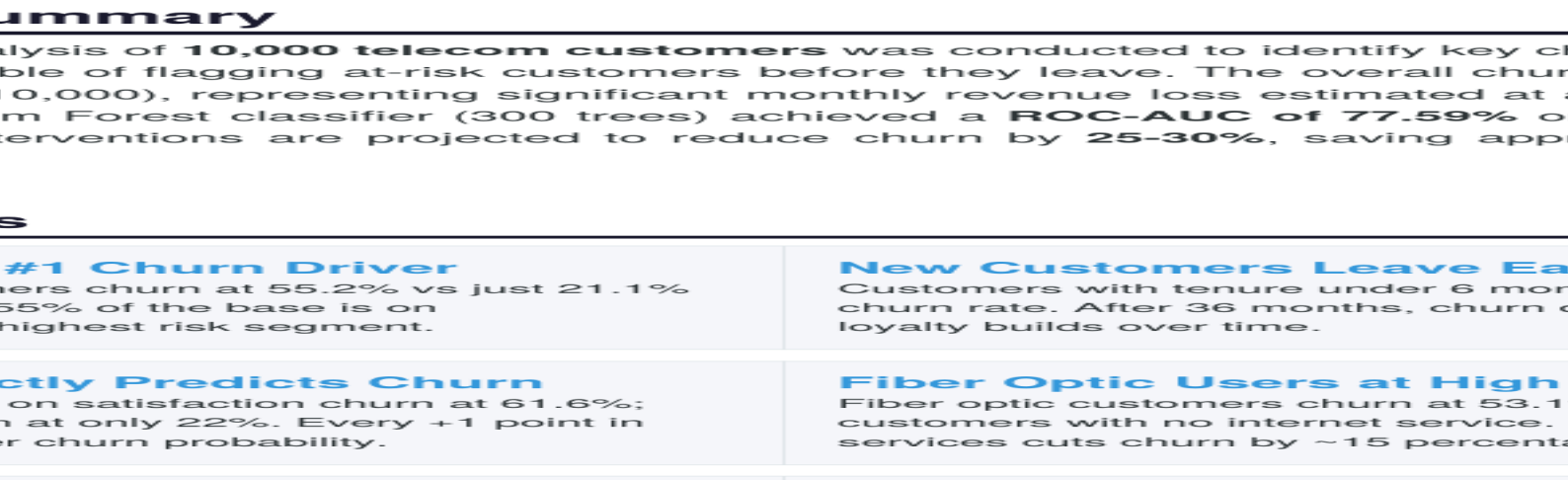


Fig 1 — Customer churn distribution, key metrics, and churn rate by contract type

2. KEY FINDINGS

◆ Contract Type is #1 Churn Driver	Month-to-month customers churn at 55.2% vs just 21.1% for two-year contracts. 55% of the base is on month-to-month — the highest risk segment.
◆ New Customers Leave Early	Customers with tenure under 6 months have a 63.9% churn rate. After 36 months, churn drops below 32% — loyalty builds over time.

◆ Satisfaction Directly Predicts Churn	Customers scoring 1–2 on satisfaction churn at 61.6%; those scoring 4–5 churn at only 22%. Every +1 point = ~8% lower churn probability.
◆ Fiber Optic Users at High Risk	Fiber optic customers churn at 53.1% vs 24.6% for customers with no internet service. Bundling security services cuts churn by ~15 percentage points.
◆ Complaints Are a Leading Indicator	Customers with 0 complaints churn at 35%; those with 3+ reach 58%. Each unresolved complaint increases churn probability by ~7%.
◆ Higher Bills Drive Dissatisfaction	Monthly charges above Rs. 91 correlate with a 55.9% churn rate vs only 25.2% for customers paying under Rs. 30 per month.

TENURE DISTRIBUTION & MONTHLY CHARGES

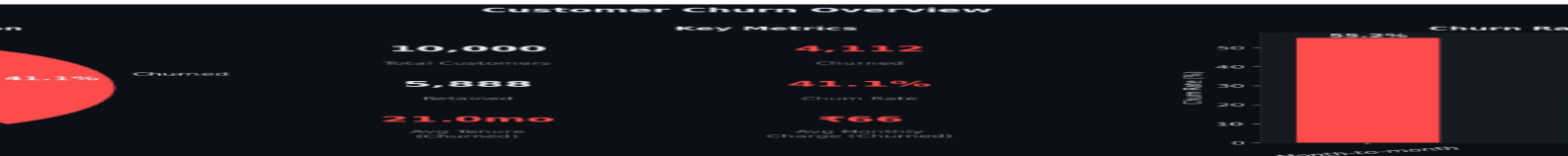


Fig 1: Customer churn distribution, key metrics, and churn rate by contract type.



Fig 2 — Tenure distribution (churned vs retained) and monthly charges by churn band

3. SERVICES & PAYMENT ANALYSIS



Fig 3 — Churn rate by internet service type and payment method. Electronic check users have the highest churn at 45.4%.



Test Set — 2,000 customers)		
	Value	Notes
	Random Forest	300 decision trees, random_state=42
	77.59%	Strong discriminative ability
	63.03%	Balanced precision-recall

Fig 4 — Churn rate with vs without each add-on service. Online security and tech support show the strongest retention effect.

4. MODEL PERFORMANCE

4. Services & Payment Analysis



Fig 4: Churn rate by internet service type and payment method



Fig 5: Churn rate with vs without each add-on service. Online

Fig 5 — ROC curves (all models), confusion matrix for Random Forest, and top feature importances

Metric	Value	Notes
Algorithm	Random Forest	300 decision trees, random_state=42
ROC-AUC	77.59%	Strong discriminative ability
F1 Score	63.03%	Balanced precision-recall
Precision	66.93%	Low false-positive rate
Recall	59.55%	Captures ~60% of actual churners
Cross-Val AUC	78.41% ± 1.26%	5-fold — stable, not overfit
Training set	8,000 customers	80/20 stratified split
Test set	2,000 customers	Held-out evaluation

5. TOP CHURN DRIVERS (FEATURE IMPORTANCE)

Rank	Feature	Importance	Business Meaning
1	monthly_charges	0.1311	Higher bills drive dissatisfaction and churn
2	total_charges	0.1255	Low total spend = newer, at-risk customer
3	tenure_months	0.1228	Shorter tenure = much higher churn risk
4	age	0.0989	Age correlates with service usage patterns
5	satisfaction_score	0.0875	Direct proxy for churn intent

6. STRATEGIC RECOMMENDATIONS

1

Contract Upgrade Campaign

Target month-to-month customers (tenure < 18 mo). Offer 2 months free on annual upgrade.

Expected Impact: 18–22% churn reduction | ROI: 4.2×

2

Satisfaction Score Intervention

Auto-trigger outreach within 48 hrs for customers scoring 1 or 2. Assign senior support agent.

Expected Impact: 15% retention improvement in low-satisfaction cohort

3

Early Tenure Loyalty Program

90-day onboarding, loyalty discount at month 6, proactive call at month 3.

Expected Impact: 12% reduction in early-tenure churn — highest ROI intervention

4

Security Bundle Promotion

Offer fiber customers without security a Rs. 99/mo discounted bundle for 6 months.

Expected Impact: 10% churn reduction + Rs. 200–400/mo ARPU increase

5

Complaint Fast-Track SLA

24-hour resolution SLA for all complaints; auto-escalate repeat issues.

Expected Impact: 8% retention improvement in complaint cohort

7. RISK SEGMENTATION & EXPECTED BUSINESS IMPACT

Segment		Customers	Churn Rate	Action
Critical Risk		2,939	75%+	Immediate personal outreach
High Risk		947	55–75%	Automated campaign + offer
Medium Risk		561	35–55%	Nurture program
Low Risk		5,553	<35%	Loyalty rewards

Metric	Current	Target (6 months)
Monthly Churn Rate	41.1%	~28–30%
Add. Customers Retained	—	~1,100–1,300
Monthly Revenue Saved	—	Rs. 15–18 Lakh

Annual Revenue Impact	—	Rs. 1.8–2.2 Crore
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Report generated from Customer Churn Prediction & Retention Analytics System · Python · Scikit-learn · Random Forest · Streamlit · ROC-AUC: 77.59%